

Set Methods Summary

Method / Property	Purpose
<code>add(value)</code>	Adds a new unique value
<code>has(value)</code>	Returns <code>true</code> if value exists
<code>delete(value)</code>	Removes a value
<code>clear()</code>	Removes all values
<code>size</code>	Returns number of values in the set

✓ 2. Array Methods Summary

Method / Property	Purpose
<code>push(value)</code>	Add to end
<code>pop()</code>	Remove from end
<code>unshift(value)</code>	Add to start
<code>shift()</code>	Remove from start
<code>splice(i, d, ...v)</code>	Add/remove at specific index
<code>slice(start, end)</code>	Returns part of array
<code>includes(value)</code>	Checks if value exists
<code>indexOf(value)</code>	Returns index of first match
<code>length</code>	Number of elements
<code>forEach(fn)</code>	Iterates over all elements
<code>map(fn)</code>	Transforms elements into new array
<code>filter(fn)</code>	Filters elements into new array
<code>sort()</code>	Sorts array in-place

✓ 3. Map Methods Summary

Method / Property	Purpose
<code>set(key, value)</code>	Adds or updates key-value pair
<code>get(key)</code>	Retrieves the value by key
<code>has(key)</code>	Checks if a key exists
<code>delete(key)</code>	Removes a key-value pair
<code>clear()</code>	Removes all key-value pairs
<code>size</code>	Number of entries in the map
<code>keys()</code>	Returns an iterator of keys
<code>values()</code>	Returns an iterator of values
<code>entries()</code>	Returns iterator of <code>[key, value]</code> pairs

Method / Property	Purpose
<code>forEach(fn)</code>	Iterates over key-value pairs

1. Array

Example Array:

js

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```
let arr = [10, 20, 30];
```

Looping Methods:

a) for loop (index-based)

js

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```
for (let i = 0; i < arr.length; i++) {  
    console.log(arr[i]); // Access via index  
}
```

b) for...of loop

js

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```
for (let value of arr) {  
    console.log(value);  
}
```

c) forEach method

js

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```
arr.forEach((value, index) => {  
    console.log(index, value);  
});
```

✓ 2. Set

Example Set:

js

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```
let s = new Set([10, 20, 30]);
```

Looping Methods:

a) for...of loop

js

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```
for (let value of s) {  
    console.log(value); // No index, just value  
}
```

b) forEach method

js

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```
s.forEach((value) => {  
    console.log(value);  
});
```

Note: `Set` doesn't support index-based access like arrays.

✓ 3. Map

Example Map:

js

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```
let m = new Map();

m.set("a", 1);

m.set("b", 2);
```

Looping Methods:

a) *for...of* loop (destructure `[key, value]`)

js

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```
for (let [key, value] of m) {

    console.log(key, value);

}
```

b) *forEach* method

js

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```
m.forEach((value, key) => {

    console.log(key, value);

});
```

c) Using *.entries()*, *.keys()*, *.values()*

js

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```
for (let key of m.keys()) {
```

```

        console.log("key:", key);
    }

    for (let value of m.values()) {
        console.log("value:", value);
    }

    for (let [key, value] of m.entries()) {
        console.log(key, value);
    }

```

Summary Table

Data Structure	Recommended Loops	Element Access
Array	for, for...of, forEach	arr[i]
Set	for...of, forEach	Only values (no index)
Map	for...of, forEach	[key, value] pairs

Let me know if you want loop examples using `while`, or nested structures like 2D arrays or Map of Sets.

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